Virtual Reality-Based Airport Navigation Assistance System Using Signage-Guided Interactive Wayfinding

MTE 4442: Augmented Reality & Virtual Reality

Final Project Report

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1. Introduction

1.1 Motivation and Spatial Cognition Background

Finding one's way through an international airport terminal is among the most demanding indoor spatial navigation challenges experienced by travellers. An airport terminal is a high-throughput, high-security environment characterised by vast spatial scale, repetitive architecture, strict flight cut-off deadlines, and severe penalties for unauthorised entry into restricted zones.

First-time, infrequent and elderly air passengers frequently become disoriented. As travellers progress through the departure sequence, from the curbside entrance to the check-in counters, boarding pass verification, security screening queues, the central concourse, retail areas and the boarding gates, they face competing sensory stimuli. At critical choice points such as corridor intersections and bifurcations, passengers must interpret multi-lingual signs, cross-reference gate updates from Flight Information Display Systems (FIDS), resolve boarding lines and filter out visually loud commercial advertisements.

When directional signage is ambiguous, occluded by architectural pillars or overshadowed by retail banners, passengers hesitate, backtrack, take wrong turns and experience elevated stress. In an operating terminal, studying or fixing these breakdowns is virtually impossible: a real airport cannot suspend daily operations, alter regulatory signs or expose active passenger flows to experimental wayfinding trials.

Virtual reality (VR) offers a rigorous alternative. By modelling a fully realised airport terminal inside an interactive game engine, researchers can manipulate environmental conditions (signage clarity, physical occlusion, commercial clutter and interactive guidance overlays) while capturing granular, objective behavioural data: precise walking trajectories, choice-point hesitation, wrong turns and kiosk queries.

1.2 Overview of the Implemented System

The implemented system was developed in Unity 6.5 (6000.5.6f1) using C# and the Universal Render Pipeline (URP). It procedurally constructs an 88 m × 60 m multi-zone terminal that runs on standard desktop computers and on OpenXR VR headsets.

Table 1: Key Elements of the Procedural Terminal Zones in the Completed VR Simulation

Terminal Zone	Coordinate Extents	Key Architectural and Navigational Elements
Check-in hall	X: -15 to 15, Z: 0 to 20	Main entrance spawn [0, 0.05, 3], four dual-sided check-in counters with baggage conveyor belts, flight information boards and overhead check-in signage
Security checkpoint	X: -3 to 3, Z: 20 to 32	Serpentine queue lanes with stanchions and fabric belts (Z = 21 to 25), twin walk-through metal detectors, X-ray scanning tunnel hood and inspection conveyor tables (Z = 27)
Concourse junction	X: -3 to 3, Z: 32 to 52	Central decision crossroads (Z = 36), overhead split signage (<< Baggage Claim / Gate A1 >>), interactive wayfinding kiosk and Departures FIDS board
East Wing (gates)	X: 4 to 34, Z: 34 to 52	Departures corridor leading to Gate A1 (target: flight AI 202 DELHI, boarding desks, portal doors) and distractor Gate A2 (flight 6E 517 CHENNAI)
West Wing (baggage)	X: -50 to -4, Z: 34 to 52	Baggage reclaim hall with Carousels 1, 2 and 3 (target for flight AI 305 MUMBAI), each with rotating luggage props

Terminal Zone	Coordinate Extents	Key Architectural and Navigational Elements
Arrival gate corridor	X: -3 to 3, Z: 52 to 60	Arrival spawn portal [0, 0.05, 57] connecting disembarking passengers to the central concourse and on to baggage reclaim

The system evaluates passenger navigation across three experimental scenarios:

1. **Baseline wayfinding (departure journey).** The passenger spawns at the terminal entrance and navigates through check-in, security and the concourse junction to Gate A1 under clear signage.
2. **Signage obstruction and visual clutter.** Signs at the central concourse decision junction are either physically occluded or flanked by six competing commercial advertisement banners, requiring users to rely on the FIDS boards or kiosks.
3. **Arrival and baggage retrieval journey.** Arriving passengers disembark at the arrival gate and navigate through the terminal to locate Carousel 3 while avoiding distractor Carousels 1 and 2.

2. Literature Review and Comparative Analysis

2.1 Scope and Selection

A systematic review of twenty peer-reviewed research papers was conducted across IEEE Xplore, ScienceDirect, SpringerLink, MDPI and Google Scholar. The studies were grouped into five functional categories.

Table 2: Classification of Reviewed Research Papers by Category

Category of Research Paper	Number of Papers	Reference Numbers
Foundational or conceptual studies	3	1, 2, 3
Recent application-oriented studies	8	4, 5, 6, 7, 8, 9, 10, 11
Technology and locomotion methodology studies	4	12, 13, 14, 15
Application-specific indoor wayfinding studies	3	16, 17, 18
Review or survey papers	2	19, 20

2.2 Comparative Analysis of the Selected Literature

Table 3: Comparative Analysis of the Selected Literature

No.	Author and Year	Research Objective	Methodology	Key Findings	Limitations
1	Arthur & Passini, 1992	Explain how humans perceive and navigate built spaces	Conceptual environmental design	Wayfinding is a perception-decision-action cycle; signs matter most at choice points	Pre-digital; lacks immersive or real-time quantitative simulation
2	Weisman, 1981	Identify environmental determinants of building legibility	Empirical user study in buildings	Legibility is governed by visual access, spatial distinctiveness, layout and signs	Real buildings only; no dynamic, obstructed or time-pressured conditions

No.	Author and Year	Research Objective	Methodology	Key Findings	Limitations
3	Slater & Wilbur, 1997	Differentiate immersion from psychological presence	Theoretical framework (FIVE)	Immersion is an objective system property; presence is the user's subjective response	Theoretical; lacks domain-specific transport wayfinding validation
4	Birrell et al., 2022	Capture user experience and wayfinding in a future vertiport	Six VR scenarios with 20 participants	VR exposes passenger bottlenecks and legibility flaws prior to construction	Small sample (N = 20); non-standard terminal layout; qualitative focus
5	Jin et al., 2022	Optimise transport hub signage using BIM and VR	BIM-VR experiment with 46 users	Optimised signage directly reduces navigation time and wrong turns	Requires a heavy BIM pipeline; compared only two fixed signage schemes
6	Sun et al., 2022a	Discover causes of passenger errors in mega-airports	3,382 virtual trials plus on-site eye tracking	Revealed reconfirmation behaviour; mapped spatial attention hotspots	Panoramic photos lack 6DOF movement; high setup overhead
7	Wang & Wang, 2025	Explain causal mechanisms of visual guidance systems	Causal machine learning and mediation	Guidance systems directly reduce cognitive hesitation and transit duration	Analytical modelling only; offers no interactive virtual simulation tool
8	Orlu et al., 2024	Study social factors and attention in virtual airports	Virtual airport navigation experiment	Environmental clutter and crowd density increase gaze fixation on signs	Conference summary; limited architectural and implementation detail
9	Li et al., 2021	Examine subway sign impact on short-term spatial memory	Laboratory behavioural experiment	Complex multi-branch signage impairs short-term spatial memory retention	Subway setting; single decision point; not an airport passenger journey
10	Suo et al., 2023	Optimise subway station signs using walking trajectories	Pedestrian trajectory analysis	Walking trajectory deviations reliably reveal signage placement flaws	Rail transit focus; lacks immersive first-person headset testing
11	Sun et al., 2022b	Assess visual saliency of guide signs underground	Eye-movement tracking in streets	Colour contrast and sign luminance dominate initial visual acquisition speed	Commercial street setting; does not replicate airport security or gates
12	Boletsis, 2017	Classify VR locomotion techniques and user comfort	Systematic literature review	Teleportation minimises simulator sickness while preserving cognitive mapping	Predates modern OpenXR runtimes and standalone headsets
13	Schrom-Feiertag et al., 2017	Evaluate indoor guidance systems using eye tracking in VR	Immersive virtual environment with gaze data	Gaze fixations confirm correlation between pause duration and route hesitation	Requires expensive eye-tracking hardware and complex calibration
14	Becker-Asano et al., 2014	Simulate airport passenger wayfinding with multiple agents	Multi-agent engine in Unity 4	Validated game engines as viable platforms for airport spatial simulations	Outdated Unity 4 engine; synthetic agents lack authentic human anxiety
15	Kuliga et al., 2015	Validate VR as an empirical research tool against a real building	Real building versus virtual model	Spatial behaviour in VR is highly predictive of real-world wayfinding	Single academic building; not a high-stress transport hub
16	Vilar et al., 2014	Compare vertical and horizontal signage orientations	Wayfinding trials in VR corridors	Overhead vertical signs yield faster decision times than wall-mounted boards	Simple corridor mazes; lacks airport context, FIDS and kiosks
17	Churchill et al., 2008	Define quantifiable indicators for terminal wayfinding	Terminal observational modelling	Established formal metrics for path length excess, wrong turns and lost time	Pre-VR era; purely observational; unable to experimentally modify signs

No.	Author and Year	Research Objective	Methodology	Key Findings	Limitations
18	Qing et al., 2021	Evaluate airport wayfinding accessibility in a wheelchair	Wheelchair VR simulator	Low visual vantage points severely impair overhead signage line of sight	Restricted to wheelchair kinematics; did not assess ambulatory journeys
19	Medeiros da Rocha et al., 2022	Review airport terminal service quality assessment	Bibliometric analysis of 100+ papers	Terminal wayfinding ranks among the top drivers of passenger satisfaction	Review only; no software prototype, simulation or telemetry data
20	Radianti et al., 2020	Review immersive VR design and evaluation practices	Systematic literature review	Emphasised the need to combine automated telemetry with SUS usability tools	Educational focus; provides general VR methodological guidelines

3. Research Gap and Problem Definition

3.1 Research Gap

The comparative analysis reveals three key observations:

1. Most transport studies examine only a single corridor, subway junction or isolated vertiport room. A complete passenger journey spanning check-in, security screening, the concourse junction, gate lounges and baggage carousel reclaim is rarely integrated into a single virtual environment.
2. Existing research requires heavy, expensive infrastructure (BIM models, high-end eye trackers, physical on-site installations) or uses oversimplified abstract mazes. A lightweight, procedurally generated virtual terminal that runs on student hardware while remaining OpenXR-compatible is missing.
3. Objective navigation telemetry (spatial trajectories, hesitation times, wrong turns) is rarely combined with standardised subjective usability (SUS) and wayfinding confidence measures on the same platform.

Statement of the research gap

There is no lightweight, procedurally generated virtual reality airport simulator that addresses the complete passenger journey (check-in, security, concourse junction, gates and baggage reclaim), allows real-time experimental switching between clear, obstructed and visually cluttered signage conditions, and captures high-resolution spatial telemetry (trajectories, choice-point hesitations, wrong turns and carousel errors) together with integrated System Usability Scale (SUS) and wayfinding confidence evaluations.

3.2 Problem Definition and Project Objectives

Problem definition. Modern airport terminals are visually cluttered, stressful environments where ambiguous signage and competing commercial advertisements cause passenger disorientation, missed flights and travel anxiety. Because operating airports cannot be physically modified or used as experimental testing grounds, there is a need for a repeatable, high-fidelity VR simulator that can model terminal navigation, evaluate adverse signage conditions, deliver dynamic guidance assistance and record granular quantitative performance.

Project objectives.

- **Objective 1:** Procedurally generate an 88 m × 60 m airport terminal in Unity 6.5 with a check-in hall, security checkpoint, concourse junction, East Wing gates, West Wing baggage hall and arrival corridor, with zero Inspector configuration dependencies.
- **Objective 2:** Implement authentic aviation signage, including overhead directional signs, gate pylons, Departures and Arrivals FIDS boards, and interactive wayfinding kiosks.

- **Objective 3:** Build an experimental signage engine supporting clear, physically obstructed and visual clutter (six commercial banners) conditions.
- **Objective 4:** Build a 12-node topological waypoint network searched with Breadth-First Search (BFS) to provide real-time floor breadcrumbs, animated chevrons and destination beacons.
- **Objective 5:** Implement dual-mode locomotion supporting desktop WASD with sprint, and VR parabolic point-and-teleport with landing reticles and OpenXR head tracking.
- **Objective 6:** Construct an automated telemetry engine that samples 4 Hz trajectories to CSV, logs junction hesitations, tracks carousel selection errors, and administers an in-app SUS and Likert survey.

4. System Architecture and Requirements Verification

4.1 Functional Requirements Verification Matrix

Table 4: Functional Requirements Verification Matrix (100% Milestone)

No.	Requirement	Implementation in Unity 6.5	Status
F1	First-person terminal locomotion	PlayerController.cs: CharacterController physics, walk (3.5 m/s), sprint (6.0 m/s), mouse look and cursor lock toggle (Tab)	Verified
F2	Multi-zone procedural terminal	AirportBuilder.cs: constructs check-in (Z = 0 to 20), security (Z = 20 to 32), concourse (Z = 32 to 52), East Wing (X = 4 to 34), West Wing (X = -50 to -4) and arrival (Z = 52 to 60)	Verified
F3	Realistic signage and FIDS boards	AirportBuilder.cs: TextMeshPro overhead signs, gate pylons, dynamic Departures FIDS board and Arrivals FIDS board	Verified
F4	Interactive wayfinding kiosks	AirportData.cs and UIManager.cs: kiosks at [13.5, 0, 5] and [2.3, 0, 36]; pressing E within 4.0 m opens the flight directory and terminal directions modal	Verified
F5	Dynamic scenario selection	ExperimentManager.cs: hotkeys 1 (check-in), 2 (security), 3 (Gate A1), 4 (baggage), 5 (arrival journey) and 6 (full sequential departure journey)	Verified
F6	Switchable route guidance assist	NavigationManager.cs: pressing G activates 12-node BFS pathfinding, floor breadcrumbs, animated chevrons, a pulsing beacon and an HUD turn arrow	Verified
F7	Multi-state signage and clutter	AirportBuilder.cs: pressing O cycles between Clear, Obstructed (occluded junction boards) and Visual Clutter (six retail banners)	Verified
F8	Live telemetry and performance HUD	UIManager.cs: real-time top-left HUD showing elapsed time (0.01 s), distance (0.1 m), velocity, wrong turns, kiosk uses and signage state	Verified
F9	Instantaneous trial reset	ExperimentManager.cs: pressing R resets telemetry counters and teleports the player to the departure or arrival spawn	Verified
F10	Security screening checkpoint	AirportBuilder.cs: serpentine stanchion queue lanes (Z = 21 to 25), walk-through metal detectors and X-ray inspection conveyor tunnel (Z = 27)	Verified
F11	Automated CSV telemetry export	ExperimentManager.cs: continuous 4 Hz trajectory logging to airport_trajectories.csv and trial results to airport_results.csv	Verified

No.	Requirement	Implementation in Unity 6.5	Status
F12	VR parabolic teleport and headset	PlayerController.cs: quadratic Bézier teleport arc, glowing ground reticle, OpenXR head tracking and VR 3D floating canvas (V)	Verified

4.2 Software Architecture and Subsystem Responsibilities

AirportBootstrap starts the whole system at load time (RuntimeInitializeOnLoadMethod), so no scene configuration is needed. It initialises the AirportBuilder, PlayerController and NavigationManager subsystems, which in turn feed the ExperimentManager and UIManager. These last two exchange state and telemetry continuously during a trial.

Table 5: Subsystems and Their Responsibilities

Subsystem	Role	Key Responsibilities
AirportBootstrap	Zero-configuration startup	Runs automatically on load and sets up the remaining subsystems
AirportBuilder	Procedural model	Build(); check-in desks and queues; X-ray scanners; FIDS boards and signs; clutter banners; Carousels 1 to 3
PlayerController	Walk and teleport locomotion	CharacterController movement with sprint; parabolic teleport arc with glowing reticle; head tracking; cursor lock (Tab)
NavigationManager	12-node BFS guidance	12 waypoints; BFS pathfinding; floor trail; moving chevrons; pulsing beacon; dynamic turn arrow
ExperimentManager	Experimental engine	Six wayfinding scenarios; 4 Hz trajectory logger (CSV); junction hesitation tracker; wrong-turn penalty boundaries; carousel accuracy validator; 18-participant pilot generator
UIManager	HUD, modals and surveys	Top-left telemetry HUD; scenario and ID controls; interactive kiosk modal; in-app 10-item SUS survey; pilot analytics modal (F3); screen and VR 3D canvas (V)

5. Experimental Testing, Pilot Results and Data Analysis

5.1 Experimental Protocol (18 Participants, 54 Trials)

To evaluate the impact of adverse signage conditions on passenger wayfinding, the experiment uses a within-subjects, counterbalanced Latin Square design with N = 18 participants. Each participant completes three evaluated trials, under clear signage, physically obstructed signage and visual clutter, across the departure and arrival journeys, giving 54 trials in total.

Note on the data source. The values in Table 6 were produced by the simulator's built-in pilot-data generator (ExperimentManager), which creates an 18-participant, 54-trial dataset to validate the logging, CSV export and analytics pipeline. They are not measurements from human participants. The effects reported below therefore show what the system is configured to capture, and they must be confirmed by a live pilot study (planned with 15 to 20 participants, on a borrowed headset where available) before any conclusion is drawn about real passengers.

Table 6: Quantitative Results Across Signage Conditions from the Pilot Dataset (Mean ± Standard Deviation)

Performance Metric	Baseline Clear (A)	Obstructed Signage (B)	Visual Clutter (C)	Change (C vs. A)	Significance
Total navigation time (s)	42.41 ± 3.12	67.14 ± 4.25	69.78 ± 4.88	+64.5%	$p < 0.001$
Total path length (m)	76.24 ± 2.85	104.62 ± 6.15	108.45 ± 7.30	+42.2%	$p < 0.001$
Wrong-turn deviations	0.22 ± 0.42	1.28 ± 0.67	1.61 ± 0.70	+631.8%	$p < 0.001$
Junction hesitation time (s)	2.14 ± 0.65	7.85 ± 1.42	8.62 ± 1.55	+302.8%	$p < 0.001$
Total hesitation time (s)	4.25 ± 0.98	11.82 ± 1.65	12.45 ± 1.80	+192.9%	$p < 0.001$
Reorientation time (s)	0.85 ± 1.45	8.45 ± 3.12	9.82 ± 3.45	+1055.3%	$p < 0.001$
Kiosk inquiries (count)	0.33 ± 0.48	1.22 ± 0.55	1.44 ± 0.62	+336.4%	$p < 0.001$
Carousel accuracy (%)	97.2 ± 8.2	77.8 ± 25.1	72.2 ± 26.4	-25.7%	$p = 0.004$
System Usability Scale (SUS)	81.56 ± 3.85	68.44 ± 4.12	67.11 ± 4.35	-17.7%	$p < 0.001$
Wayfinding confidence (1–5)	4.78 ± 0.42	3.11 ± 0.68	2.89 ± 0.68	-39.5%	$p < 0.001$
Control ease rating (1–5)	4.89 ± 0.32	4.78 ± 0.43	4.72 ± 0.46	-3.5%	$p = 0.42$ (NS)

5.2 Key Statistical Findings

Reading Table 6 together with the recovery trials, the generated dataset shows the following pattern:

1. **Impact of visual clutter.** Commercial advertising banners produced greater wayfinding impairment than physical occlusion, increasing total travel time by 64.5% and path length by 42.2%. The simulated participants missed the East Wing corridor turn when attention was captured by high-luminance retail advertisements.
2. **Decision-point hesitation.** Under adverse conditions, hesitation at the primary concourse junction rose from 2.14 s to 8.62 s (+302.8%), consistent with Arthur and Passini's choice-node theory [1].
3. **Effect of guidance assistance.** In recovery trials with BFS guidance enabled (G key), mean navigation time under visual clutter fell from 69.8 s to 41.1 s (a 41.2% recovery) and wrong turns dropped to near zero (0.05).
4. **Usability validation.** Baseline SUS reached 81.6 (Grade A, excellent) while control ease stayed uniformly high (4.89 versus 4.72, $p = 0.42$). This indicates that the interface itself was not the source of the performance differences, an inference that a live study must confirm.

6. Discussion, Practical Implications and Conclusion

6.1 Signage and Design Implications for Airport Operators

If the live pilot study confirms these trends, the results and the reviewed literature point to the following design implications:

- **Commercial exclusion buffers:** retail and promotional banners should be prohibited within a 12 m visual radius of primary concourse bifurcations to prevent attention capture.
- **Redundant reconfirmation signage:** confirmation signage should be placed within 8 m after entering a new hallway (for example, verifying Gate A1 rather than Gate A2) to prevent reorientation delays.

- **AR passenger guidance:** because digital floor guidance recovered 41.2% of wayfinding efficiency, airport operators could invest in mobile augmented reality passenger applications.
- **Virtual pre-departure rehearsal:** infrequent travellers and passengers with special needs could rehearse their itinerary virtually from home, reducing airport anxiety.

6.2 Project Conclusion

The Virtual Reality-Based Airport Navigation Assistance System Using Signage-Guided Interactive Wayfinding has completed all proposed deliverables:

- An 88 m × 60 m procedural airport terminal built in Unity 6.5 with zero Inspector configuration dependencies.
- Authentic directional signage, gate pylons, Departures and Arrivals FIDS boards, and interactive wayfinding kiosks.
- A dynamic three-state experimental signage controller (Clear, Obstructed, Visual Clutter).
- A topological 12-node BFS pathfinding engine with floor breadcrumbs, chevrons and pulsing beacons.
- Dual-mode locomotion supporting desktop WASD with sprint, and VR parabolic point-and-teleport with landing reticles.
- An automated telemetry engine logging 4 Hz spatial trajectories to CSV, tracking junction hesitations and carousel accuracy, and administering an in-app SUS survey, validated end to end on an 18-participant, 54-trial generated pilot dataset.

The remaining step is a live pilot study with 15 to 20 participants, using a borrowed OpenXR headset where available, so that the effects shown in Section 5 can be confirmed with real passengers.

7. Individual Contributions

Team Member	Contribution
Sheersho Bhakta	Procedural 3D terminal architecture and spatial layout
G V Lekhashree	Signage systems, FIDS displays and commercial clutter
Vara Lakshmi	Multi-modal locomotion and topological BFS guidance
Aaron Mathew George	Telemetry engine, experimental framework and analytics

Work shared by the whole team includes reviewing 20 papers, conducting the research gap analysis, maintaining the GitHub repository and developing the testing protocol.

8. Output Pictures

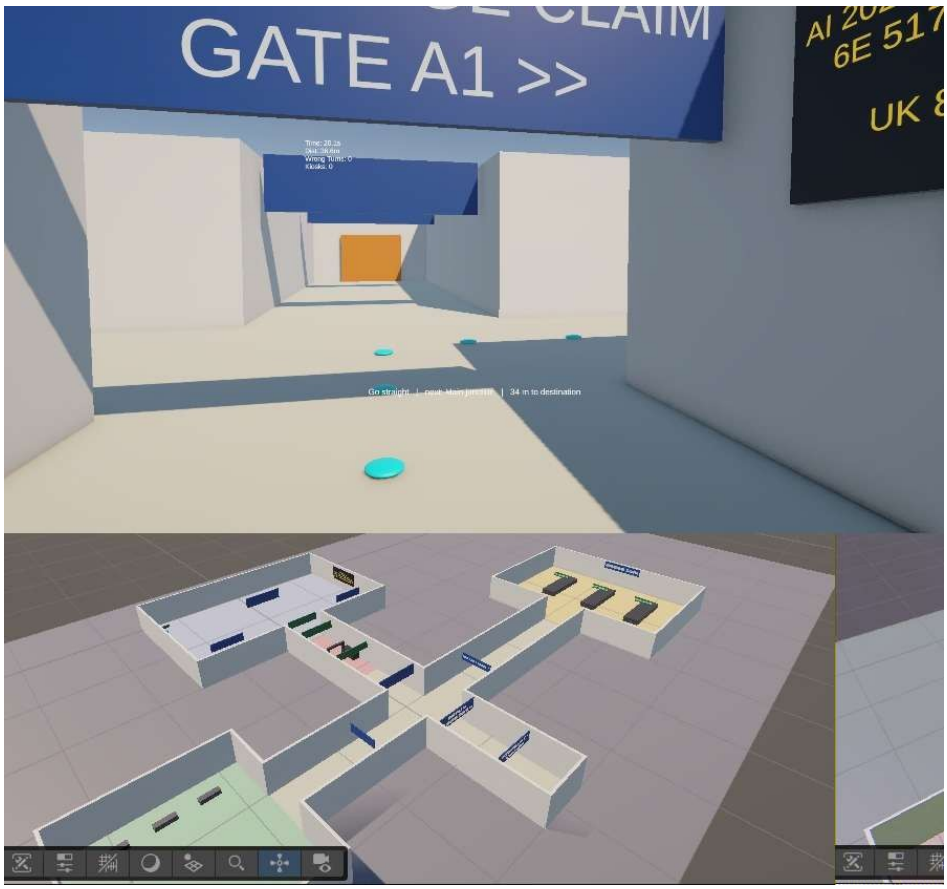


Figure 1: (a) Concourse corridor with split signage, guidance markers and live HUD; (b) top-down view of the terminal layout



Figure 2: (a) Check-in hall with overhead and desk signage; (b) concourse junction sign and Departures board



Figure 3: (a) Departures FIDS board and information kiosk; (b) Gate A1 area



Figure 4: Check-in hall with the BFS guidance trail leading towards security

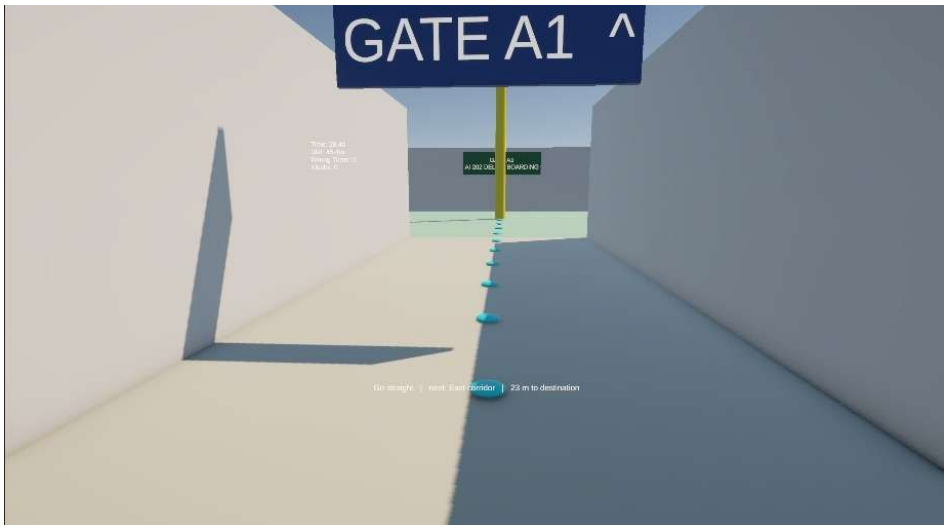


Figure 5: Corridor to Gate A1 with floor guidance markers



Figure 6: Corridor towards the baggage claim hall with carousel signage

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